Awwal Oladipupo

PhD Candidate in Chemical Engineering and Scientific Computing

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Research Interests

Leveraging computational techniques, including machine learning, atomistic simulations, multi-scale modeling, and software engineering, to accelerate the discovery and design of novel materials for energy and environmental applications.

Education

University of Michigan, United States — Aug 2023 – Apr 2028

- PhD in Chemical Engineering and Scientific Computing — CGPA 4.00/4.00
- Master's in Chemical Engineering — Aug 2023 – May 2024; CGPA 4.00/4.00

University of Lagos (UNILAG), Nigeria — Oct 2015 – Nov 2021

- B.Sc. in Petroleum and Gas Engineering
- Valedictorian of Faculty/College of Engineering and Top 1% at the Institution (First Class Honors, 4.93/5.00)
- Scholaro CGPA 3.97/4.00
- Capstone Project: Modelling of liquid loading in gas wells using a software-based approach

Honors / Awards / Prizes

- Best Oral Presentation Award and Cash Prize for Track 2 Chemical Engineering Symposium Research Presentation: Material Design and Process Optimization, University of Michigan — Sept 2026
- Rackham Conference Travel Grant 26–27, University of Michigan — Sept 2026
- Michigan Institute for Computational Discovery and Engineering (MICDE) fellowship — Aug 2026
- Promoted to Full Member, Sigma Xi, The Scientific Research Honor Society — Feb 2026
- Nominated for Phi Kappa Phi for being part of the top scholars at the University of Michigan — Feb 2026
- Rackham Conference Travel Grant 25–26, University of Michigan — Sept 2025
- The 2025 Visionary Award, LLMs for Materials and Chemistry Hackathon — Sept 2025
- Elected Member, Tau Beta Pi, The Engineering Honor Society — Aug 2025
- Associate Member, Sigma Xi, The Scientific Research Honor Society — Aug 2025
- Engineering Graduate Symposium 3-minute thesis competition People's choice award, University of Michigan — Feb 2025
- Best Engineering Graduate Award, Nigeria Society of Engineers — Dec 2022
- Google Africa Developer Scholarship — Mar 2022
- Invited Scholar, Stanford Exposure to Research and Graduate Education program (SERGE) — Oct 2022
- EducationUSA Scholar, Opportunity Funds Program — Jul 2022
- University of Lagos Dean Award (best graduating student in Faculty/College of Engineering) — Jan 2022

- University of Lagos Head of Department Award (best graduating student in Petroleum and Gas Engineering) — Jan 2022
- The Nigeria Society of Engineers (NSE) Academic Excellence Award (Ikeja Branch) — Aug 2022
- The Oxford and Cambridge Club of Nigeria (OxBridge) Leadership Award — Jan 2022
- New Era Foundation Prize — Jan 2022
- ECPM Engineers Award — Jan 2022
- Abu Bakri Shonibare Memorial Prize — Jan 2022
- Adeniyi Akanni Engineering Award — Aug 2021
- University of Lagos Endowment Award — Jan 2017 & Jan 2020
- CASOR UNILAG Academic Excellence Award — Apr 2019 & Apr 2020
- Most Dependable Student Award, Society of Petroleum Engineering (Lagos) — Dec 2021
- Dr. Olumide Philips Scholarship (DROPS) Award — Aug 2020
- Society of Petroleum Engineers Lagos Section Academic Excellence Award — Aug 2018 & Aug 2019

Research Grants and High-Performance Computing Allocations

- **Principal Investigator**, ACCESS Explore Award (CHM260051), *TurboChIMES: A Multilayered Machine-Learned Interatomic Model for Increased Simulation Efficiency* (Mar. 2026–Mar. 2027). Awarded 400,000 ACCESS Credits for research computing on the Stampede3 supercomputer to develop and validate machine-learning interatomic potentials for accelerated molecular simulations.
- **Principal Investigator**, ACCESS Computing Allocation (CHM250017), *Evaluating Transferability of Multilayered Machine-Learned Interatomic Potentials* (Feb. 2025–Feb. 2026). warded 400,000 ACCESS Credits for computational benchmarking of machine-learning interatomic potentials, evaluation of transferability across chemical systems, and assessment of simulation accuracy and computational efficiency using Stampede3 and Ranch storage.

Research Experience

Graduate Research Assistant — Oct 2023 – Apr 2028

The Lindsey Lab, University of Michigan

Machine Learning and Simulation for Accelerated Materials Design and Discovery

- Develop physics-informed machine-learning methods and computational frameworks to investigate atomistic phenomena and accelerate materials discovery through predictive molecular simulations.
- **TurboChIMES: Multilayered Machine-Learned Interatomic Potentials.** Developed and implemented TurboChIMES to improve the representation of short- and long-range atomic interactions at reduced model complexity, achieving a reported 1,500% increase in computational speed and a 90% reduction in computational cost for molecular dynamics simulations. This work is associated with DOE-funded research on advanced energy materials and computational materials discovery.
- **Hierarchical Transfer Learning.** Co-developed an enhanced hierarchical transfer-learning framework for machine-learned interatomic potentials to improve transferability across chemical systems and reduce the computational demands of model development. Contributed to DOE-supported research in multiscale materials modeling and computational materials discovery.
- **Alchemical Machine-Learned Interatomic Potentials.** Develop an alchemical framework for parametrically linear machine-learned interatomic potentials to accelerate model training, improve transferability, and reduce computational cost.

- **Agentic AI for ChIMES.** Develop an agentic AI framework to automate force-field development, validation, optimization, and molecular simulation workflows.
- **Collaborative DARPA-Funded Energetic Materials Research ($2 million).** Responsible for the computational modeling efforts investigating light-sensitive polymer strategies for controlling energetic-material sensitization and detonation, developing computational frameworks to characterize material response under energetic loading.
- **Aluminum-Oxygen Propulsion Technology.** Develop computational frameworks to investigate and support the design of aluminum-oxygen (Al-O) propulsion technologies through atomistic modeling and simulation.

Related Federally Funded Research Projects

- **Formulation Engineering of Energy Materials via Multiscale Learning Spirals.** U.S. Department of Energy (DOE), $660,000, Oct 2021–Sep 2024. Principal Investigator: Prof. Rebecca K. Lindsey; Lead Principal Investigator: Prof. Michael Tsapatsis. Contributed to ChIMES-based computational modeling of perovskite formation and nucleation to investigate processing–structure–performance relationships in advanced solar materials.
- **Membrane-Catalyst Co-Design for Transformative Manufacturing.** U.S. Department of Energy (DOE), $525,000, Oct 2023–Sep 2025. Contributed to computational modeling research using ChIMES-based methods to support DOE objectives in advanced manufacturing.

Lead Developer, MaterialSim AI Agent Sept 2025

LLM Hackathon for Applications in Materials and Chemistry

- Led a cross-disciplinary team in developing MaterialSim, an AI agent integrating large language models with computational materials science tools to automate molecular dynamics workflows.

Lead Developer 2026–Present

EZRA: Easy Zero-Setup Routing Assistant

Open-Model Agents for Materials Science, Chemistry, and High-Performance Computing

- Designed and developed EZRA, an open-source toolkit that enables researchers to download and run open-weight language models locally, configure research agents, and transfer workflows to high-performance computing (HPC) clusters.
- Implemented workflows for scientific document preparation, local agent execution, and job staging on SLURM-managed computing infrastructure, helping researchers move from local experimentation to cluster-based execution.
- Developed a modular agent architecture supporting custom system prompts, preferred language models, and domain-specific configurations for materials design, chemistry research, and synthesis planning.
- GitHub: https://github.com/Awwal41/EZRA

Graduate Research Assistant Pro Bono (Part-time) Dec 2021 – Aug 2023

University of Lagos

Methanolysis of Ram-fat over Modified Kaolinite-clay Impregnated Calcium-oxide from Chicken-eggshell: Optimization Study by Response Surface Methodology and Meta-heuristic Algorithms

- Optimized the methanolysis of ram-fat over modified kaolinite-clay impregnated calcium-oxide from chicken-eggshell by minimizing the biodiesel yield using particle swarm optimization (PSO), genetic algorithm (GA), and firefly algorithm (FA) developed codes.

Sand Production Prediction and Control using Artificial Intelligence

- Developed and coded machine and deep learning algorithms using Python to predict sand production in the field and proffer the best control scheme based on prediction.

Undergraduate Researcher Mar 2020 – Nov 2021

University of Lagos

Modelling of liquid loading in gas wells using a software-based approach

- Developed artificial neural network and Bayesian neural network deep learning models and optimization algorithm codes using Python and MATLAB, respectively, for liquid loading prediction.

Publications

1. Aritra Roy, Kevin Shen, Andrew MacBride, **Awwal Oladipupo**, et al.; Ben Blaiszik*. *Knowledge Infrastructure: Better ways to know things in the Age of Large Language Models.* (Submitted to *Digital Discovery*).
2. Andrew MacBride, **Awwal Oladipupo**, Aritra Roy, Kevin Shen, et al.; Ben Blaiszik*. *Action Systems: New Ways to Do Things in the Age of Large Language Models.* In prep (to be submitted to *Digital Discovery*).
3. **Awwal Oladipupo**, Vallabh Vasudevan, Akhila Ponugoti, Toheeb Balogun, Jodie Yuwono. *MaterialSim: An AI Agent For Automating Molecular Dynamics Simulations and Materials Analysis with Natural Language and Machine Learning.* Submitted to JOSS.
4. **Awwal Oladipupo**, Benjamin Laubach, Sayed Almohri Rebecca Lindsey. *TurboChIMES: A Multi-layered Machine-Learned Interatomic Model Targeting Improved Description of Short and Long-Range Interactions with a scale-up in simulation efficiency.*doi: chemrxiv.15007194. Submitted to Journal of Chemical Therory and Computation.
5. **Awwal Oladipupo**, Dean Sweeney, Roshini Dantuluri, Bryan R. Goldsmith, and Rebecca K. Lindsey. *Bridging film and droplet for predicting liquid loading in gas wells with machine learning.* In prep.
6. Aritra Roy, Kevin Shen, Andrew MacBride, **Awwal Oladipupo**, et al.; Ben Blaiszik*. *From Knowledge to Action: Outcomes of the 2025 Large Language Models (LLM) Hackathon for Applications in Materials Science and Chemistry.*;doi:.
7. Lindsey R, **Oladipupo A**, Bastea S, Steele B, Kuo I-FW, Goldman N. *Hierarchical Transfer Learning: An Agile and Equitable Strategy for Machine-Learning Interatomic Models. npj Computational Materials.* 2026. doi: 10.1038/s41524-025-01863-4.
8. Mepaiyeda B, Ezeh M, Olafadehan O, **Oladipupo A**, Adebayo O, Osaro E. *Integration of Machine Learning and Feature Analysis for the Optimization of Enhanced Oil Recovery and Carbon Sequestration in Reservoirs.* ChemRxiv. 2025; doi:10.26434/chemrxiv-2025-5v32t. Submitted to the *Journal of Petroleum Science.*
9. **Oladipupo A**, Johnson I, Adebayo O, Abhulimen K, Theophilus F. *Leveraging Data-driven Approach for Sand Production and Management in Oil and Gas Wells. Petroleum Science and Engineering* 10.11648/j.pse.20250902.13.
10. E. O Ajala, A. B Ehinmowo, **A. D Oladipupo**, S Opawoye, A. M Ndana, D. A Ariyoosu. *Experimental Study and Computational Modelling for the Optimisation of Biodiesel Production from Waste Ram Fat Using Modified Kaolinite-clay Impregnated Calcium-oxide Catalyst. Nigerian Journal of Technological Development.* .
11. Mepaiyeda, E. B.; Oluwayomi, I. **A.; Oladipupo**, A. D.; Odutola, T. O. *Prediction of Gas Hydrate Formation Temperature in Pipelines using Artificial Neural Network (ANN) and Firefly Algorithm (FA). Petroleum & Coal.* 2023, Vol. 65 Issue 3, p824–835.
12. Abhulimen, K.E., Abhulimen, K.E. & **Oladipupo, A.D.** *Modelling of liquid loading in gas wells using a software-based approach. J Petrol Explor Prod Technol* (2022). doi:10.1007/s13202-022-01525-x

Articles

1. Awwal Oladipupo. *Agentic AI and the Physics of Teamwork: Lessons from Resistors in Series and Parallel.* Medium.
2. Awwal Oladipupo. *Is Cloud Computing Really "Cloud" or Just Remote Computer Services?* Medium.
3. Awwal Oladipupo. *The effects of scaling weights in generating matrix solutions: Potential effects on ML algorithms.* Medium.
4. Awwal Oladipupo. *Intro: Application of Graph Theory for Understanding Chemical Systems.* Medium.
5. Awwal Oladipupo. *Importance of Android Apps Testing.* Medium.

Presentations

The presenting author is underlined.

- Awwal D. Oladipupo, Thomas Sundberg and Rebecca K. Lindsey. *Exploring Alchemical Compression Pathways for Reducing Elemental Combinatorial Explosion In Machine Learned Interatomic Potential.* American Institute of Chemical Engineers (AIChE) Conference, Minnesota, USA. — Nov 2026
- Awwal D. Oladipupo and Rebecca K. Lindsey. *Multi-Layer Architecture for Machine-Learned Interatomic Models for Efficient Large-Scale Simulations.* University of Michigan Chemical Engineering Graduate Symposium, Ann Arbor, Michigan, USA. — Sept 2026
- Awwal D. Oladipupo. *Accelerating Molecular Simulations with Multilayer Machine-Learned Interatomic Potentials.* Michigan Institute for Computational Discovery and Engineering (MICDE), PhD student seminar series, Ann Arbor, Michigan, USA. — Sept 2026
- Awwal D. Oladipupo and Rebecca K. Lindsey. *Enhancing Efficiency of Machine-Learned Interatomic Models through a Multilayered Strategy.* American Institute of Chemical Engineers (AIChE) Conference, Boston, USA. — Nov 2025
- Awwal D. Oladipupo and Rebecca K. Lindsey. *Designing an architectural framework of machine-learned interatomic potentials for increased efficiency for exascale simulation.* University of Michigan Chemical Engineering Graduate Symposium, Ann Arbor, Michigan, USA. — Sept 2025
- Awwal D. Oladipupo and Rebecca K. Lindsey. *Modification of Machine Learned Interatomic Potential Design Architecture to Increase Simulation Efficiency.* Midwest Thermodynamics and Statistical Mechanics Conference, Madison, Wisconsin, USA. — Jun 2025
- Awwal D. Oladipupo. *Multilayered machine learned model for fast design and discovery of functional materials.* University of Michigan Engineering Graduate Symposium 3-minute thesis competition, Ann Arbor, Michigan, USA. — Feb 2025
- Awwal D. Oladipupo and Rebecca K. Lindsey. *Evaluating the Performance of a New Transfer-Learning Strategy for Understanding Materials Using ChIMES Machine Learning Interatomic Potential.* University of Michigan Chemical Engineering Graduate Symposium, Ann Arbor, Michigan, USA. — Sept 2024
- Awwal D. Oladipupo and Rebecca K. Lindsey. *Evaluating Efficacy of a New Transfer-Learning Strategy for Efficient Parameterization of the ChIMES Machine Learning Interatomic Potential.* Midwest Thermodynamics and Statistical Mechanics Conference, Ann Arbor, Michigan, USA. — Jun 2024

Work Experience

Graduate Student Instructor Fall 2026

CHE 466: Process Dynamics and Control, University of Michigan

- Support undergraduate instruction in process dynamics and control, reinforcing students' understanding of dynamic process behavior, feedback control, and the analysis of chemical engineering systems.
- Hold office hours to clarify course concepts, address student questions, and support problem-solving in process dynamics and control.
- Grade homework assignments and examinations, evaluating student understanding and application of core chemical engineering principles.

Tutor (Part Time) Jan 2021 – Aug 2023

Light Education Consultancy

- Tutored undergraduate students both in Nigeria and the United Kingdom (UK) virtually on Chemical and Petroleum engineering courses, including Fluid Mechanics, Solid Mechanics, Thermodynamics, Gas Dynamics, Optimization of energy processes, Mathematical Modelling and Analysis, and application of data science and machine learning to energy systems.

Associate Android Developer Feb 2023 – Aug 2023

BabbanGona

- Contributed to the development of an Android-based platform designed to help smallholder farmers optimize farm land use and manage agricultural produce more efficiently.

Mobile Developer Intern Feb 2022 – Feb 2023

BabbanGona

- Supported the development and maintenance of Android applications focused on agriculture and rural empowerment.
- Learned agile development practices while contributing to mission-driven technology in a high-impact startup.

Production and Field Engineering Trainee Aug 2019 – Jan 2020

Summit Oil International

- Participated in work-over operation in Otien-1: de-completion, maintenance of Christmas tree valves, and extensive fishing. Followed all operational and procedural activities at the site and managed workload given by engineers in the field in compliance with HSE and operational standards. Carried out weekly presentations to engineers in the field regarding the operations carried out weekly.

Leadership / Volunteer Experience

Professional Service 2025 – Present

Peer Review, Judging, and Academic Service

- Volunteer Judge, Open Scientific Intelligence (OSI) Hackathon (Oct. 2026–Present). Evaluating projects developed by researchers and domain experts applying and advancing innovative scientific intelligence technologies, including LLM-enabled tools and AI-driven approaches for scientific discovery and research.
- Volunteer Judge, Seventh Annual Switch Energy Alliance (SEA) Global Energy Competition (Sept 2026–Present).
- Reviewer, Journal of Artificial Intelligence Research (JAIR) (Feb 2026–Present). Review areas include machine learning robustness and generalization, conversational AI, large language models, agentic AI systems, and empirical evaluation of large-scale neural architectures. Completed one review assignment. ‘‘
- Reviewer, American Chemical Society (ACS) Journals (Feb. 2026–Present). Completed six peer-review assignments evaluating manuscripts authored by researchers affiliated with institutions worldwide,

spanning chemistry, materials science, scientific computing, and AI-driven modeling.

- Reviewer, Pre-Publication Support Service (PREPSS) (Sept 2025–Present).
- Ad Hoc Reviewer, International Journal of Artificial Intelligence in Engineering Fields (IJAIEF) (Jul 2025–Present). Selected by the Editor-in-Chief to provide expert manuscript evaluations.
- Peer Reviewer, International Journal of Materials Science and Applications (IJMSA) (Jul 2025–Jul 2028). Completed one review assignment.
- Judge, CODiE Awards (Jul 2025–Present). Served as one of approximately 400 experts evaluating innovative technology products across approximately 103 categories.
- Volunteer Judge, Sixth Annual Switch Energy Alliance (SEA) Global Energy Competition (Sept 2025–Nov 2025). Evaluated innovative solutions addressing global energy poverty submitted by 285 student teams from 27 countries.
- Poster Judge, Chemical Engineering Undergraduate Research Symposium, University of Michigan (Apr 2025). Evaluated undergraduate research posters for scientific merit, clarity, and presentation quality.
- Volunteer, EducationUSA Lagos Graduate Fair (Sept 2022). Facilitated event operations and participant registration for over 500 attendees.
- Member, The Industry Discourse (TID) Planning Committee (2017–2021). Coordinated engagement with industry professionals to strengthen connections between academia and the energy sector.

Leadership, Recruitment, and Community Engagement 2020 – Present

Academic and Professional Organizations

- **Organizer**, Open Scientific Intelligence (OSI) Hackathon 2026 (Oct. 2026–Present). Helped organize an international hackathon advancing open scientific intelligence, bringing together researchers, developers, and innovators to build LLM-enabled tools and AI-driven technologies for scientific discovery. Supported an initiative that attracted over 1,000 sign-ups and 105 project submissions, with entries progressing to the judging phase.
- **Peer Mentor, Department of Chemical Engineering, University of Michigan** (Aug. 2026–Present).
  - Mentor first-year PhD students on navigating graduate school, developing effective research practices, and transitioning into independent research.
  - Guide students in preparing and maintaining research lab notebooks, organizing research activities, and identifying and connecting with faculty whose expertise aligns with their research interests.
  - Support students in planning for their candidacy examinations by helping them understand expectations, organize their preparation, and develop effective strategies for academic and research progress.
- **Junior Member, Leadership Team and Liaison to the Director**, Center for Simulation and Data Intensive Research in Chemical Engineering (CSiDIR-CHE), University of Michigan (Feb. 2026–Present). Serve as a liaison between junior members and the Center Director, supporting communication, coordinating community-building initiatives, and facilitating interdisciplinary engagement and knowledge exchange among students, postdoctoral researchers, research staff, and faculty working at the intersection of physics-based simulation, scientific computing, and machine learning.
- **Intelligence Infrastructure Architect**, Eleaktra: Intelligent Pipeline Monitoring and Leak Detection, (Jan 2022–Present)
  - Lead the design of the project's superintelligence infrastructure, developing the software and intelligence architecture for integrating machine learning, sensor data, anomaly detection, and decision-support capabilities into an intelligent pipeline-monitoring platform.
  - Design the architecture to support multi-sensor data fusion, automated identification and classification of abnormal operating conditions, leak localization, and the delivery of actionable insights for infrastructure operators.

  - Apply expertise in machine learning, scientific computing, engineering modeling, and software development to extend established computational leak-detection capabilities toward an extensible intelligent infrastructure platform.
  - Consider architectural pathways for deployment across oil and gas pipelines, municipal water networks, mining operations, and other fluid-transport systems, with potential integration into existing industrial monitoring and control environments.
- **Team Lead and Developer (Project-Alpha)**, CALL-E: Your Code Is Calling Hackathon, 2026
  - Led a three-person team, including myself, in developing an AI-powered platform integrating large language models with CALL-E's voice-agent infrastructure to extend AI capabilities beyond text-based interactions.
  - Coordinated technical development and integration of AI models with CALL-E to support voice-enabled agent workflows, enabling models to interact with real-world phone-based tasks.
  - Contributed to the design and implementation of a developer-oriented platform for connecting AI models with phone-call automation and tracking agent interactions and execution outcomes.
  - Submission: https://devpost.com/software/project-alpha-8u2p91
- **Co-chair**, Engineering Graduate Symposium 2026 (Oct 2025–Feb 2026). Led programmatic planning and execution of the college-wide symposium in collaboration with the Office of the Associate Dean for Graduate and Professional Education (ADGPE).
- Member, Graduate Recruitment Committee, Chemical Engineering, University of Michigan (2023–2025). Developed and implemented initiatives to attract prospective graduate students to the department.
- **Assistant Programs Chair**, Team Havilah (Jan. 2020 – Aug. 2023). Mentored more than 30 students, conducted outreach to over 500 students, and provided education on climate change, sustainability, career development, and academic success.
- Chemical and Petroleum Engineering Peer Tutor, University of Lagos (Oct. 2017– Nov.2021). Taught and mentored students across chemical, petroleum, and gas engineering courses.
- Team Lead / Assistant, Various Engineering Competitions (Aug. 2017– Nov. 2020), including:
  - PETMAT Professor Awojobi Design Competition Team (PAADC), 2017
  - Society of Petroleum Engineers (SPE) Nigeria PetroQuiz, 2020
  - European Association of Geoscientists and Engineers (EAGE) competitions and related student activities

Technical Skills

- Programming Languages: Python, Kotlin, C++, SQL, typescript, javascript, and bash.
- Scientific Software Development: Object-Oriented Programming (OOP), software architecture, API development, algorithm design, debugging, testing, and performance optimization
- Machine Learning & AI: Machine learning, optimization algorithms, scientific AI, data-driven modeling, and AI agent development
- Scientific Computing: Numerical methods, high-performance computing (HPC), parallel computing, and simulation workflow automation
- Data Analysis & Visualization: Power BI, Tableau, Minitab, data processing, and statistical analysis
- Development Tools: Git, GitHub, Jira, Linux, and Microsoft Office Suite (Word, Excel, PowerPoint)

Society / Activities

- Elected Member, Tau Beta Pi, The Engineering Honor Society — Aug 2025 – Present
- Full Member, Sigma Xi, The Scientific Research Honor Society — Aug 2025 – Present
- Member, American Institute of Chemical Engineers (AIChE) — Apr 2025 – Present

- Member, Chemical Engineering Graduate Society (ChEGS) — Aug 2023 – Present
- Member, Black in AI — 2022 – Present
- Member, Society of Petroleum Engineers (SPE) — 2017 – Present
- Member, European Association of Engineers and Geo-scientist (EAGE) — 2019 – Present